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Adult ADHD-Related Poor Quality of Life: Investigating the Role of Procrastination

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ABSTRACT

The link between Attention Deficit and Hyperactivity Disorder (ADHD) and reduced quality of life (QoL) has been well established. The current study examines the role of procrastination in explaining this link, providing a new focus for research and therapy. This study examines the involvement of procrastination in accounting for ADHD-related reduced QoL. Adult participants ($N = 132$) completed an online survey consisting of validated scales to assess ADHD symptoms, procrastination levels, and QoL. An indirect pathway between ADHD and quality of life through procrastination was examined. Higher levels of ADHD symptoms correlated with higher procrastination and lower quality-of-life scores. Indirect pathways between ADHD symptoms and poor QoL through levels of procrastination were identified. These results shed further light on ADHD and its association with reduced QoL and account for this link by the negative impact of procrastination on day-to-day functioning. Future research is warranted to design effective interventions for consumers with ADHD-related procrastination, targeting different aspects of quality of life.

1 | Introduction

Attention-deficit/hyperactivity disorder (ADHD) is a prevalent neurodevelopmental condition characterized by persistent patterns of inattentive, hyperactive, and impulsive behaviors, resulting in functional impairment (American Psychiatric Association 2013; Faraone et al. 2015). Longitudinal studies indicate that a substantial portion of individuals diagnosed with ADHD during childhood continue to exhibit symptoms into adulthood (Song et al. 2021).

Adults with ADHD often experience significant impairments that diminish their overall quality of life (QoL) across various life stages (Agarwal et al. 2012; Joseph et al. 2019). QoL, in this context, refers to individuals' subjective sense of well-being and their ability to function effectively in different life domains (Karimi and Brazier 2016), such as physical and psychological health, social relationships, and satisfaction with the living environment (The Whoqol 1998). As defined by Felce and Perry (1995), QoL encompasses an individual's objective and subjective evaluations of physical, material, social, and emotional well-being, along with personal development and purposeful activity, all influenced by their values (p. 60). Enhanced QoL correlates with the fulfillment of basic physiological and security needs, perceived physical and mental health, and a sense of purpose and meaning in life (Karimi and Brazier 2016). Impaired QoL associated with ADHD often manifests in areas such as work productivity, social relationships, psychological well-being, and physical health (Brod et al. 2005).

What explains the link between adult ADHD and reduced QoL? Various studies have examined the potential mechanisms underlying this association. Research has demonstrated that the persistence of ADHD into adulthood predicts lower QoL (Yang

Summary

- Procrastination's significance: Recognize procrastination as a key factor influencing the link between ADHD and reduced quality of life.
- ADHD symptom correlation: Higher ADHD symptoms correlate with increased procrastination and lower quality-of-life scores, underscoring the need to address procrastination in ADHD assessments.
- Intervention focus: Target procrastination in ADHD management plans to enhance overall well-being and day-to-day functioning.
- Future research emphasis: Stay informed about evolving research on effective interventions for ADHD-related procrastination and its impact on quality of life.

et al. 2013) and that symptom reduction following medication use has been linked to improvements in QoL (Weiss et al. 2010). Several studies have reported that depressive and anxiety symptoms (Nadeau et al. 2015; Seo et al. 2014; Zhang et al. 2021; Hennig et al. 2017; Matthies et al. 2018), as well as maladaptive cognitions and emotional distress (Pan et al. 2023), mediate the relationship between ADHD symptoms and life satisfaction/QoL. Additionally, self-reported deficits in executive functioning (Zhang et al. 2021), lack of social support (Hennig et al. 2017), and adverse childhood experiences (Matthies et al. 2018) have been identified as contributing factors to the diminished QoL observed in adults with ADHD. The present paper aims to investigate another potential individual difference that may further explain the reduced QoL associated with ADHD, namely, procrastination.

Procrastination, defined as the voluntary but irrational delay of intended actions (Steel 2007), is a prevalent tendency experienced by almost everyone at some point (Harriott and Ferrari 1996). Similar to ADHD, procrastination impacts various life domains, including academic performance, health behavior, personal finance, and work outcomes (Kljajic and Gaudreau 2018; Sirois 2021; Gamst-Klaussen et al. 2019; Metin et al. 2018; Nguyen et al. 2013). The psychological toll of procrastination has also been established. For instance, procrastination has been linked to distress and poor mental health (Rice et al. 2012; Stead et al. 2010), negative feelings about the self such as shame (Fee and Tangney 2000; McCown et al. 2012), lower self-image and self-efficacy (Van Eerde 2003), low self-compassion (Sirois 2014), as well as poor mood regulation (Sirois and Pychyl 2013). This heavy psychological toll explains why precrastination has been shown to adversely affect QoL and life satisfaction (Beutel et al. 2016; Berber Çelik and Odaci 2022). Notably, procrastination has also been linked to poor physical health, further reducing individuals' QoL (Sirois 2016).

Looking into the link between procrastination and ADHD, Ferrari (2000) found no such association among typically developed adults. However, in a subsequent study co-authored by the same researcher, adults with ADHD reported significantly higher rates of procrastination in different life domains (Ferrari and Sanders 2006). In a study focusing on college students, Miller (2007) found higher rates of procrastination among students with ADHD relative to the average. In light of the mental-health dimensional model, ADHD lends itself to a conceptualization as a continuous trait (Coghill and Sonuga-Barke 2012), thereby elucidating the detrimental effect of ADHD symptoms on QoL in a broad context, regardless of an existing diagnosis. Correspondingly, a positive correlation between procrastination and ADHD symptoms was demonstrated in several studies (Niermann and Scheres 2014; Ashworth and Mccown 2018; Altgassen et al. 2019; Bolden and Fillauer 2019; Bodalski et al. 2023; Netzer Turgeman and Pollak 2023).

Given the predictive roles of both procrastination and ADHD on poor QoL, alongside the observed link between ADHD and procrastination, it is plausible that ADHD-related procrastination is an important contributor to poor QoL in different domains. This study aims to investigate the role of procrastination in explaining the relationship between ADHD and QoL. Specifically, it was hypothesized that ADHD symptoms would be associated with higher procrastination and lower QoL across different domains, that procrastination would be associated with lower QoL, and that the link between ADHD and lower QoL would be accounted for by higher procrastination.

2 | Method

2.1 | Participants and Procedure

The study received approval from the departmental ethics committee (approval no. 31082022). The survey was conducted using Amazon Mechanical Turk (MTurk), an online crowdsourcing platform commonly utilized for experimental research purposes (Paolacci and Chandler 2014). Unlike many previous studies employing student samples, we opted to recruit a more diverse adult sample through Amazon MTurk. Recent meta-analytic findings have indicated that, on the whole, mean scores and variances in MTurk samples closely align with those obtained through other means across a broad spectrum of measures (Keith et al. 2022). Between February 13th and 22nd, 2022, individuals aged 18 and above registered on Mturk were invited to participate in the study and were compensated \$3 based on the estimated time required to complete the survey. Informed consent was obtained from all participants.

To determine the necessary sample size, we utilized a Monte Carlo simulation-based tool developed by Schoemann et al. (2017) for calculating power in path analysis. For detecting indirect effects in a single mediation model with a 99% confidence level, a power of 0.8, and moderate-to-large correlations between variables in the model (r = 0.40), a sample size of n = 130 was deemed necessary.

Data were collected from a total of 201 participants. Among these, 69 individuals were excluded from the analysis: 44 completed less than 70% of the survey, 23 failed to correctly answer two attention check questions, and 2 completed the survey in less than 3 min. Consequently, responses from N = 132 participants were included in the analysis. Participants' ages ranged from 24 to 60 years (Mage = 39.9, SD = 9.3); 73 were male, 58 were female, and one identified as other. Regarding educational attainment, 24 (18%) participants were high school graduates, 44 (33.5%) were undergraduate students, 33 (25%) held a bachelor's degree, and 31 (23.5%) held a master's degree. Seven participants reported a diagnosis of ADHD, with three of them having taken medication on the day of the survey.

2.2 | Measures

2.2.1 | Quality of Life

Two scales were utilized to assess QoL. The primary instrument employed was the Adult ADHD Quality of Life Questionnaire (AAQoL) (Brod et al. 2005). This scale was chosen due to its emphasis on evaluating the impact of ADHD across four domains: productivity, life outlook, relationships, and psychological health. Comprising 29 5-point scale items, the scale generates a total score derived from all 29 items, with higher scores indicative of better QoL. Research has demonstrated that the AAQoL exhibits strong internal consistency ($\alpha = 0.93$) and test–retest reliability (interclass correlation coefficient [ICC] = 0.86), and it effectively discriminates between individuals with and without ADHD (Brod et al. 2005; Matza et al. 2007, 2011).

As a complementary measure of QoL, the World Health Organization WHOQOL-BREF Quality of Life Assessment (BREF) (The Whoqol 1998) was employed. This scale was selected alongside the AAQoL, despite its lack of specificity to ADHD, as it encompasses additional dimensions such as physical health and environmental resources, which are not covered by the AAQoL. The BREF consists of 5-point items targeting four domains: physical health (7 items), psychological health (6 items), social relationships (3 items), and environment (8 items). Note that the official manual of the BREF does not include a total score across the four domains, and therefore, each domain was analyzed separately. Scores for each domain were computed, with higher scores indicating better QoL. The BREF has demonstrated good internal consistency, ranging from $\alpha = 0.91$ to $\alpha = 0.93$ (Whoqol Group. 1998).

2.2.2 | Procrastination

The Irrational Procrastination Scale (IPS) features nine items measuring the degree of irrational delay on a 5-point scale. The English original version and the Hebrew translation have yielded a good internal consistency, with Cronbach's $\alpha = 0.91$ and 0.87, respectively (Steel 2010; Netzer Turgeman and Pollak 2023). A mean score was computed, with higher scores representing higher levels of procrastination.

2.2.3 | ADHD Symptoms

The Adult ADHD Self-Report Scale (ASRS-V1.1) (Kessler et al. 2005) stands as a frequently employed dimensional tool for assessing current ADHD symptoms. Comprising 18 items corresponding to ADHD symptoms, each item is evaluated based on frequency using a 5-point scale. Kessler et al. (2005) reported a high level of internal consistency for the questionnaire, registering at 0.88. Additionally, Kessler et al. (2005) noted a sensitivity of 68.4 and a specificity of 99.6% for the scale. Given that the original version of the questionnaire features an item directly addressing procrastination, it was adapted by omitting this specific item, resulting in the ASRS-R. Consistent with the dimensional conceptualization of ADHD (Coghill and Sonuga-Barke 2012), the disorder was approached as a continuous trait within the study. Mean scores were computed, with higher scores denoting elevated levels of ADHD symptoms.

2.2.4 | Covariates

The Kessler Screening Scale for Psychological Distress (K6) (Kessler et al. 2002) consists of six questions measuring the frequency of psychological distress on a 5-point scale, ranging from "all of the time" to "none of the time." A total score was computed, with higher scores representing lower levels of distress. The scale has been widely validated in cross-cultural contexts (Kessler et al. 2010). The measure was included in the analysis as a covariate to control for the variance in the outcome measures that is explained by psychological distress.

A demographic questionnaire required participants to provide background information. Age, gender, and education level were used as covariates in the mediation models.

2.3 | Data Analysis

Statistical analysis was performed on standardized scores. We conducted normality tests for each dependent variable. Descriptive statistics, reliability, and inter-correlation analyses for the model variables were computed.

The hypotheses that ADHD symptoms are associated with higher procrastination and lower QoL, and that procrastination is linked to lower QoL, were tested using Pearson correlations. The hypothesis that procrastination explains ADHD-related reduced QoL was examined through regression-based mediation analyses utilizing the PROCESS macro (Hayes 2017) for SPSS.

For mediation analyses, we conducted a missing values analysis, which revealed that three variables (BREF physical health and social relationship, K6) contained missing values, and 0.55% of the total values incorporated in the analyses were missing. In order to avoid loss of power and biased estimates, the missing values were imputed by the expectation–maximization (EM) method using SPSS version 29 statistical software.

The mediation analyses examined the associations between the ASRS, IPS, BREF, and AAQoL scores, with a specific focus on the direct and indirect pathways between the ASRS and AAQoL scores. We assessed significance using bias-corrected bootstrapping with 5000 resamples and calculated 95% confidence intervals.

TABLE 1 | Descriptive Statistics for the study's variables.

Variables	*M*	SD	Skewness	Kurtosis	Cornbach's α
AAQoL					
Total	67.2	19.9	−0.20	−0.74	0.95
BREF					0.94
Physical Health	72.9	18.5	−0.79	0.30	0.81
Psychological Health	65.4	22.3	−0.75	−0.05	0.87
Social Relationship	64.8	27.1	−0.9	0.07	0.87
Environment	73.0	17.4	−0.77	0.49	0.86
ASRS-R					
Mean score	2.2	0.9	0.73	−0.35	0.96
IPS					
Mean score	2.4	0.9	0.46	−0.29	0.90
K6					
Total score	8.0	7.5	0.94	−0.31	0.95

Abbreviations: AAQoL, Adult ADHD Quality of Life; ASRS-*R*, The Adult ADHD Self Report Scale revised to exclude an item targeting procrastination; BREF, World Health ganization's Quality of Life Instrument—short form; IPS, irrational procrastination scale; K6, Kessler Psychological Distress Scale.

TABLE 2 | Inter-correlations between the study variables.

Variable	1	2	3	4	5	6	7
1. AAQoL	—						
2. BREF Physical Health	0.60	—					
3. BREF Psychological	0.49	0.65	—				
4. BREF Social relationship	0.41	0.46	0.77	—			
5. BREF Environment	0.50	0.63	0.70	0.62	—		
6. IPS	−0.67	−0.57	−0.56	−0.45	−0.51	—	
7. ASRS-R	−0.73	−0.51	−0.31	−0.24	−0.33	0.72	—
8. K6	0.46	0.33	0.39	0.30	0.36	−0.36	−0.30

Note: All correlations are significant at the $p < 0.01$ level.
Abbreviations: AAQoL, Adult ADHD Quality of Life; ASRS-*R*, The Adult ADHD Self Report Scale revised to exclude an item targeting procrastination; BREF, World Health ganization's Quality of Life Instrument—short form; IPS, irrational procrastination scale; K6, Kessler Psychological Distress Scale.

3 | Results

3.1 | Descriptive Statistics

Table 1 presents means, standard deviations, normal distribution indicators, and Cronbach's α reliability measure for all the main variables in the study.

3.2 | Correlation Analysis

Bivariate Pearson correlations are presented in Table 2. ADHD symptoms (the revised ASRS score excluded an item targeting procrastination, as stated above) were positively and strongly linked to procrastination (IPS), so higher ADHD levels were related to higher procrastination scores. ADHD symptoms were found to correlate negatively and strongly with scores on all the domains of QoL and with the psychological distress score (a higher K6 score indicates lower psychological distress). The procrastination scores were negatively and strongly correlated with scores on all the domains of QoL and with psychological distress.

3.3 | Mediation Analyses

We examined the role of procrastination in explaining the relationship between ADHD symptoms and QoL. QoL was measured using two scales, one general (the total AAQoL score) and four specific domains of QoL (the physical health, psychological health, social relationships, and environment scales of the BREF). All models were controlled for psychological distress, age, education, and gender. The first model, which predicts the overall AAQoL scores, showed that ADHD symptoms positively predicted procrastination, which in turn negatively predicted the AAQoL score. A 5000-resampled bootstrapping confidence interval (CI) estimating the indirect effect indicated that this pathway was significant ($B = -0.21$, SE = 0.08, 95% CI [−0.38, −0.07]). Accordingly, higher self-reported ADHD symptoms predicted lower levels of QoL through higher procrastination (see Figure 1).

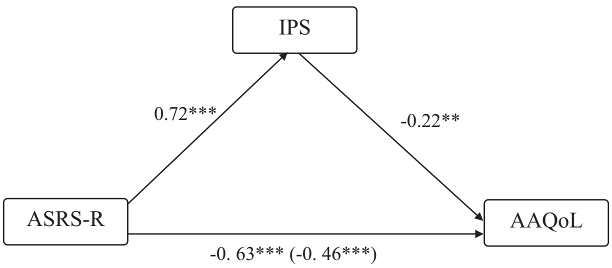

FIGURE 1 | Path analysis predicting AAQoL total score. ASRS-*R*, The Adult ADHD Self-Report Scale (revised), IPS, The Irrational Procrastination Scale, AAQoL, Adult ADHD Quality of Life Questionnaire. Coefficients are standardized. The model controls for K6, age, education, and gender. The indirect effect: −0.16, SE = 0.06, 95% CI [−0.28, −0.05]. $^{**}p < 0.01$ $^{***}p < 0.001$.

Four mediation models examined if and to what extent procrastination mediates the relationship between ADHD symptoms and the four subscales of BREF. As presented in Figure 2A–D, procrastination fully explained the relationship between ADHD symptoms and physical health (indirect effect: $B = -0.29$, SE = 0.13, 95% CI [−0.54, −0.05]), psychological health (indirect effect: $B = -0.58$, SE = 0.13, 95% CI [−0.83, −0.31]), social relationships (indirect effect: $B = -0.62$, SE = 0.17, 95% CI [−0.96, −0.28]), and environment (indirect effect: −0.39, SE = 0.12, 95% CI [−0.63, −0.16]). In all four models, the negative and significant effect of ADHD symptoms on BREF became small and insignificant after including the procrastination score; therefore, it can be concluded that the link between ADHD symptoms and BREF is fully explained through procrastination.

4 | Discussion

The study examined the following hypotheses: ADHD symptoms would be associated with higher procrastination and lower QoL across different domains, procrastination would be associated with lower QoL, and the link between ADHD symptoms

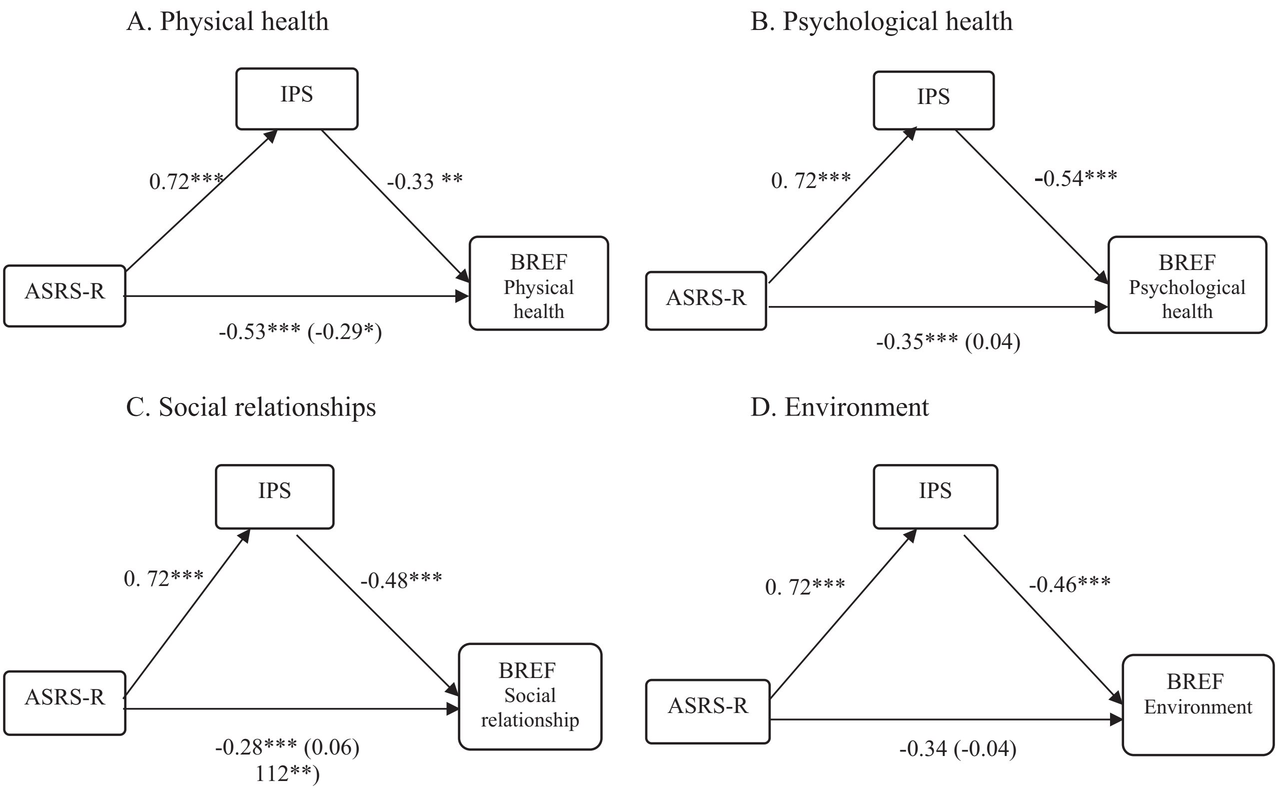

FIGURE 2 | Path analysis predicting the BREF domain scores. ASRS-*R*, The Adult ADHD Self-Report Scale (revised), IPS, The Irrational Procrastination Scale, BREF = the World Health Oganization's Quality of Life Instrument—Brief Form. The model controls for K6, age, education, and gender. Coefficients are standardized. The indirect effects were the following. Physical health domain −0.2382, SE = 0.10, 95% CI [−0.44, 0.04]; Psychological health domain −0.39, SE = 0.09, 95% CI [−0.56, −0.21]; Social relationships domain −0.34, SE = 0.09, 95% CI [−0.52, −0.16]; Environment domain −0.34, SE = 0.11, 95% CI [−0.55, −0.14]. $^{*}p < 0.05$ $^{**}p < 0.01$ $^{***}p < 0.001$.

and lower QoL would be accounted for by higher procrastination. The findings supported all these hypotheses.

We found a positive correlation between ADHD symptoms and procrastination, as well as a negative correlation between ADHD symptoms and QoL. Procrastination emerged as a significant mediator in the relationship between ADHD symptoms and overall QoL, as well as between ADHD symptoms and specific domains of QoL, namely, physical and psychological health, social relationships, and satisfaction with the living environment. All five mediation models yielded a consistent result whereby it is suggested that higher levels of procrastination constitute a mechanism through which higher ADHD potentially leads to reduced QoL across domains.

Revealing the role of procrastination in explaining the link between ADHD and QoL adds to other known mediating factors, such as depressive/anxiety symptoms (Nadeau et al. 2015; Seo et al. 2014; Zhang et al. 2021; Hennig et al. 2017; Matthies et al. 2018), poor executive function (Zhang et al. 2021), and lack of social support (Hennig et al. 2017). It is possible that some of these variables overlap, as emotion regulation problems, lower executive functions, and procrastination often coincide (Bodalski et al. 2023; Bolden and Fillauer 2019). Nevertheless, our findings suggest that procrastination should be recognized and targeted for improving the QoL of people with ADHD.

The link found in this study between ADHD and procrastination is well established, and several studies examined possible mechanisms underlying it. Altgassen et al. (2019) demonstrated that the link between ADHD symptoms and procrastination was mediated by the performance of everyday prospective memory tasks. Bolden and Fillauer (2019) identified executive functions as a mediating factor in this link. Bodalski et al. (2023) partially accounted for the relationship between ADHD and procrastination through difficulties in emotion regulation and reduced self-esteem. Finally, the link between ADHD and procrastination was partially mediated by low expectancy for completing the task and higher sensitivity to delay (Netzer Turgeman and Pollak 2023). In the context of the current research, understanding these mechanisms may inform researchers and clinicians of strategies for reducing procrastination in people with high levels of ADHD symptoms and thereby improving their QoL. These studies, together with the results of the current paper, suggest that ADHD and procrastination share common cognitive and motivational mechanisms and that these mechanisms are risk factors for lower QoL. Indeed, poor executive functions were found to be a risk factor for poor QoL in aging (Davis et al. 2010; Gamage et al. 2018) and in various health conditions (e.g., Schworer et al. 2020; Stern et al. 2017; Sharfi and Rosenblum 2016; Roy et al. 2021). Similarly, lower emotion regulation has been linked to poor QoL in several physical and mental disorders (Wilmer et al. 2021; Barberis et al. 2017; Durosini et al. 2022).

QoL was assessed utilizing two established self-report measures, AAQoL and WHOQOL-BREF, designed to measure the levels of QoL among people with ADHD and the general population, respectively. The WHOQOL-BREF assesses QoL in four life domains separately: physical health, psychological health, social relationships, and the living environment. Importantly, ADHD symptoms and procrastination were associated with poor QoL in each of the domains. While the literature on ADHD has consistently established the negative impact of ADHD on nearly all life domains (for review, see Faraone et al. 2021), the impact of procrastination on QoL has been less exhaustive. However, it is important to note that the literature on the detrimental impact of procrastination has not been limited to academic and vocational success and emotional health but also to physical health and social outcomes (for review, see Sirois 2016; Giguère et al. 2016). The results of the current study suggest shared mechanisms for the pervasive impact of ADHD and procrastination on people's QoL.

On both the AAQoL and WHOQOL-BREF, the correlations found between self-reported ADHD symptoms and QoL emerged as negative and significant, and more so for AAQoL (see Table 2). This difference may be explained by the AAQoL's specificity in addressing the QoL challenges of people with ADHD, whereas the BREF is a general health-related QoL scale. Notably, this difference corresponds to the wording of the items in the two questionnaires: In the BREF questionnaire, all the questions except for one are formulated positively (for example: "How satisfied are you with yourself?"), while in the AAQOL questionnaire, the wording is primarily negative (for example: "During the past two weeks, how troubled have you been by not having quality time to spend with others?"). As the correlation between two negatively-worded scales is often higher than the correlation between positively- and negatively-worded scales due to common-method variance (Podsakoff et al. 2003), it is plausible that the negatively-worded ADHD scale would correlate more strongly with the AAQoL than with the BREF.

The above pattern merits serious consideration as to the effect of wording. Wording effects are demonstrated in logically inconsistent answers to regular and reversed items of the same construct (Kam 2016). Such effects are thought to arise from response biases such as carelessness, compliance, and comprehension difficulty (Baumgartner et al. 2018; Weijters et al. 2013). In the case in point, the disparity in the answers raises the issue regarding the influence of the way items are phrased on the findings in this and other studies of individuals with ADHD symptoms. There is a possibility that people with ADHD are more sensitive to the response biases exemplified above. Notably, the Adult ADHD Self-Report Scale (ASRS) questionnaire, commonly used to assess ADHD symptoms, is formulated with predominantly negative wording. As we could not find studies on this subject, we call for further research into the wording effect regarding ADHD.

5 | Limitations

Given the cross-sectional design of our study, other interpretations of the results are possible, for instance, that poor QoL might increase procrastination and aggravate ADHD symptoms. The investigation into the temporal directionality of the relationships among ADHD, procrastination, and QoL awaits future longitudinal research endeavors. In addition, the data were obtained from self-reports. One could assume that using reports of friends or partners may have contributed to a better picture of each participant's symptoms and behaviors. However, prior research suggests that adults with ADHD tend to provide more accurate self-reports of their symptoms compared to reports from their partners (Kooij et al. 2008).

6 | Implications

The results of the current study highlight the relationship between ADHD, procrastination, and low QoL. While longitudinal and intervention studies are warranted to validate these findings, our results suggest that mitigating procrastination among adults with ADHD could improve their QoL. Various intervention programs have demonstrated efficacy in reducing procrastination across different settings (for review, see Kachgal et al. 2001; van Eerde and Klingsieck 2018; Zacks and Hen 2018), offering potential help for individuals with ADHD. Moreover, some implemented and researched intervention programs were tailored to address procrastination specifically for consumers with ADHD (e.g., Ramsay 2020; Safren 2006). Such evidence indicates that targeted, evidence-based interventions that reduce procrastination tendencies may be a powerful tool in improving the QoL of individuals with ADHD and that more research is needed to design such strategies and analyze their impact.

Author Contributions

Both authors substantially contributed to the conception and design of the work, as well as the writing of the manuscript. R.N.T. collected and analyzed the data.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author, Ruth Netzer Turgeman, upon reasonable request. We are committed to transparency and open communication regarding our research data.

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